

House Price Prediction Analysis – Key Findings

Project Overview

The objective of this project was to analyze housing data and build machine learning models to predict house prices based on different property features. The analysis helped identify the factors that have the greatest impact on house prices and evaluate the performance of predictive models.

Data Preparation

- Loaded and explored the housing dataset.
- Checked for missing values and found that the dataset was complete.
- Converted categorical variables (such as air conditioning, parking preferences, and furnishing status) into numerical format using one-hot encoding.
- Selected relevant features for price prediction.
- Split the dataset into 80% training data and 20% testing data.

Exploratory Data Analysis

- A histogram of house prices showed that most properties fall within a moderate price range, while a few houses have significantly higher prices.
- The correlation heatmap helped identify relationships between different features and house prices.
- House area showed the strongest positive relationship with price.

Important Factors Affecting House Price

The analysis revealed that the following features have the greatest influence on house prices:

1. **Area** – Larger houses generally have higher prices.
2. **Bathrooms** – Houses with more bathrooms tend to be more valuable.
3. **Stories** – Multi-story houses often have higher market value.
4. **Parking** – Additional parking spaces positively affect property prices.
5. **Air Conditioning** – Houses with air conditioning are usually priced higher.
6. **Preferred Area** – Properties located in desirable areas command higher prices.

Model Performance

Two machine learning models were trained and compared:

1. Linear Regression

- Simple and easy-to-interpret model.
- Provided a good baseline for house price prediction.

2. Random Forest Regressor

- Captured more complex relationships between features.
- Produced more accurate predictions compared to Linear Regression.

The models were evaluated using:

- **MAE (Mean Absolute Error)** – Measures average prediction error.
- **RMSE (Root Mean Squared Error)** – Measures overall prediction accuracy.
- **R² Score** – Indicates how well the model explains price variation.

Overall, the Random Forest model achieved better performance and provided more accurate predictions.

Key Insights

- House price is influenced by more than just property size.
- Features such as air conditioning, parking availability, and preferred location significantly impact property value.
- Some houses with similar areas had different prices, showing that amenities and location play an important role in pricing.

Recommendation

Real estate businesses should focus on properties that offer valuable amenities such as parking, air conditioning, and desirable locations. Highlighting these features in marketing campaigns can increase buyer interest and improve property value.

Conclusion

This analysis demonstrated how data analytics and machine learning can be used to predict house prices and identify the factors that influence property values. Among all features, area, location, and amenities were the most important drivers of house prices, while the Random Forest model delivered the best prediction performance.